

HPC AI Observability and Security

Architecture Reference | Open Source Sovereign Stack

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Four architecture diagrams:

1

HPC Cluster

Physical topology: compute nodes, NVLink, InfiniBand, storage, scheduler

2

Training Step Timeline

What happens inside one step: data load, forward, backward, AllReduce

3

Observability Stack

Collection, storage, and visualization pipeline layer by layer

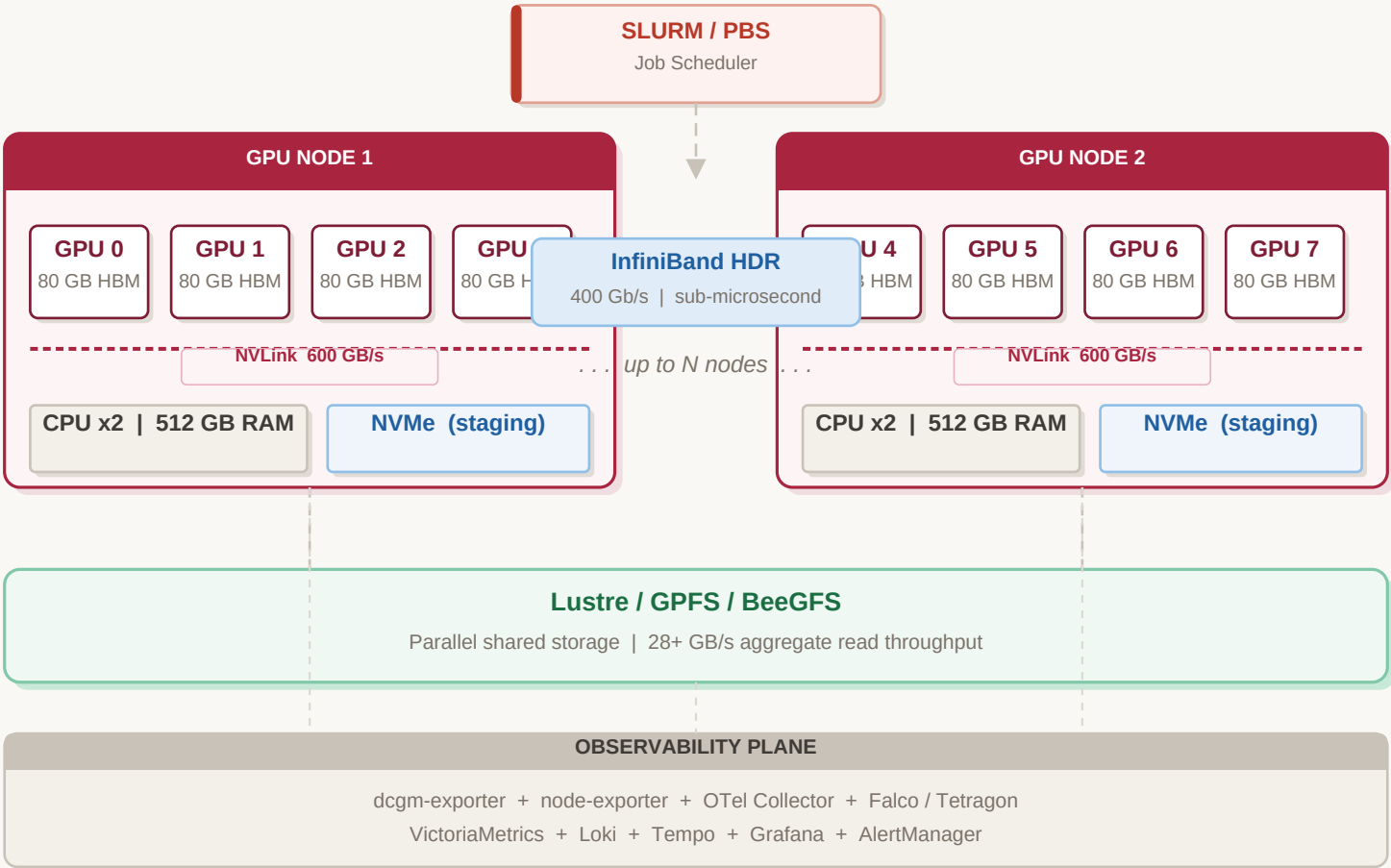
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Security Threat Surface

Attack vectors specific to HPC AI and their detection methods

Diagram 1 | HPC Cluster Architecture

Full system view: compute, network, storage, scheduling, observability

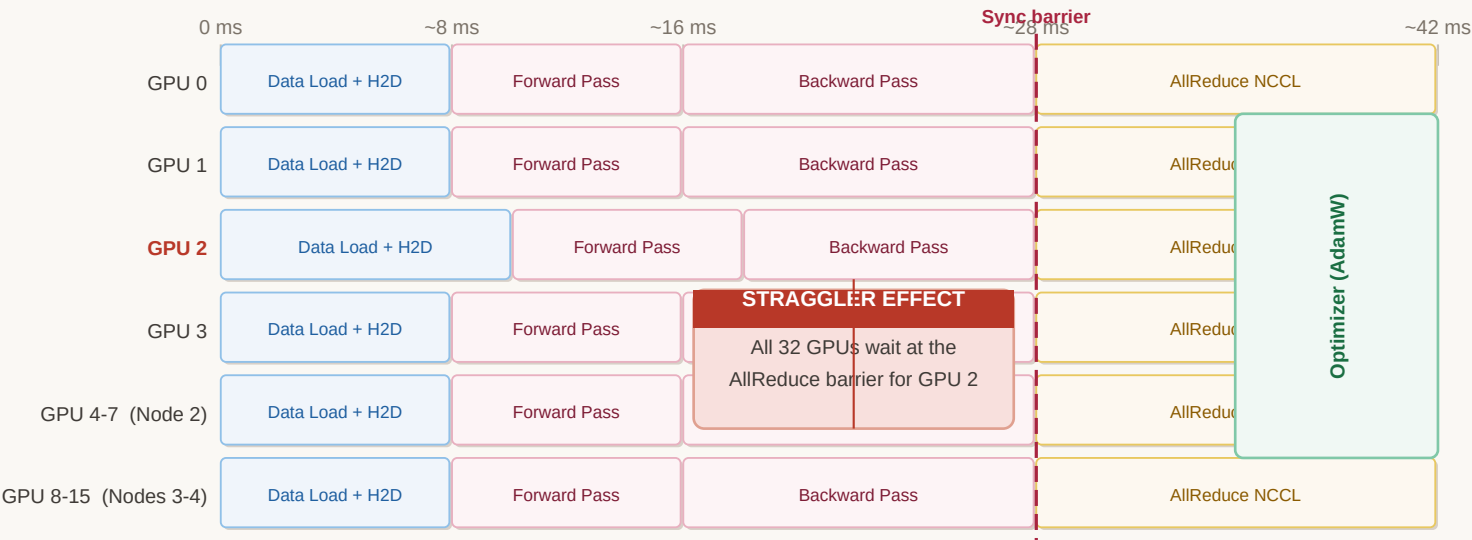


- GPU: SM utilization, HBM bandwidth, ECC errors, clock throttle
- Network: IB port counters, retransmit rate, per-link throughput
- Storage: Lustre throughput, metadata latency
- Model: loss, gradient norm, step time, AllReduce latency

Diagram 2 | Distributed Training Step Timeline

One training step across 4 nodes / 32 GPUs | Total duration ~42 ms

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Key insight: GPU utilization reported at 87%. Effective compute: ~19%. AllReduce alone: 43% of step time.

I/O + Memory Transfer GPU Compute Network Comm. (NCCL) Optimizer

Diagram 3 | Observability Stack Architecture

Sovereign open source pipeline: collection, storage, visualization, alerting

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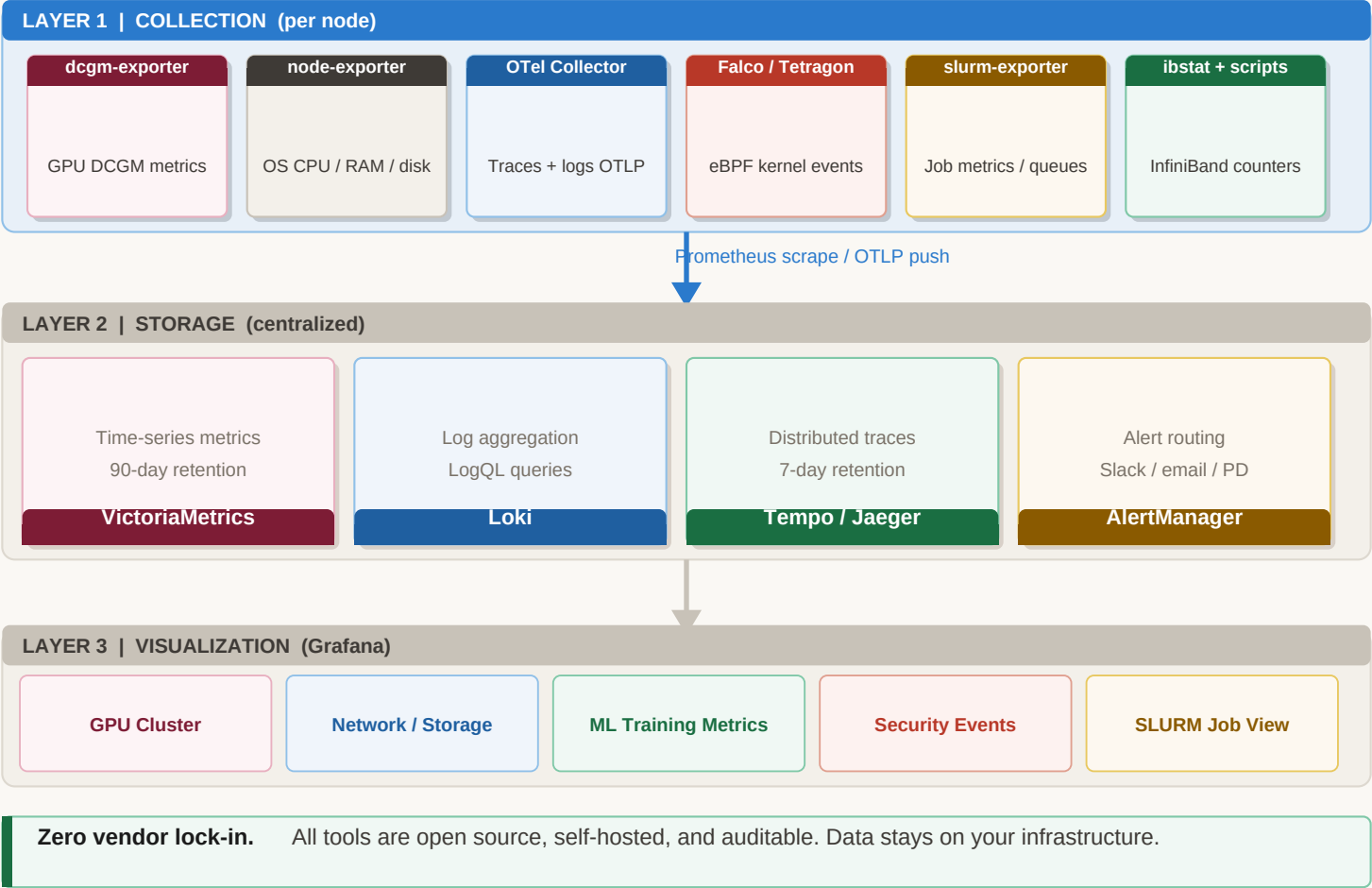
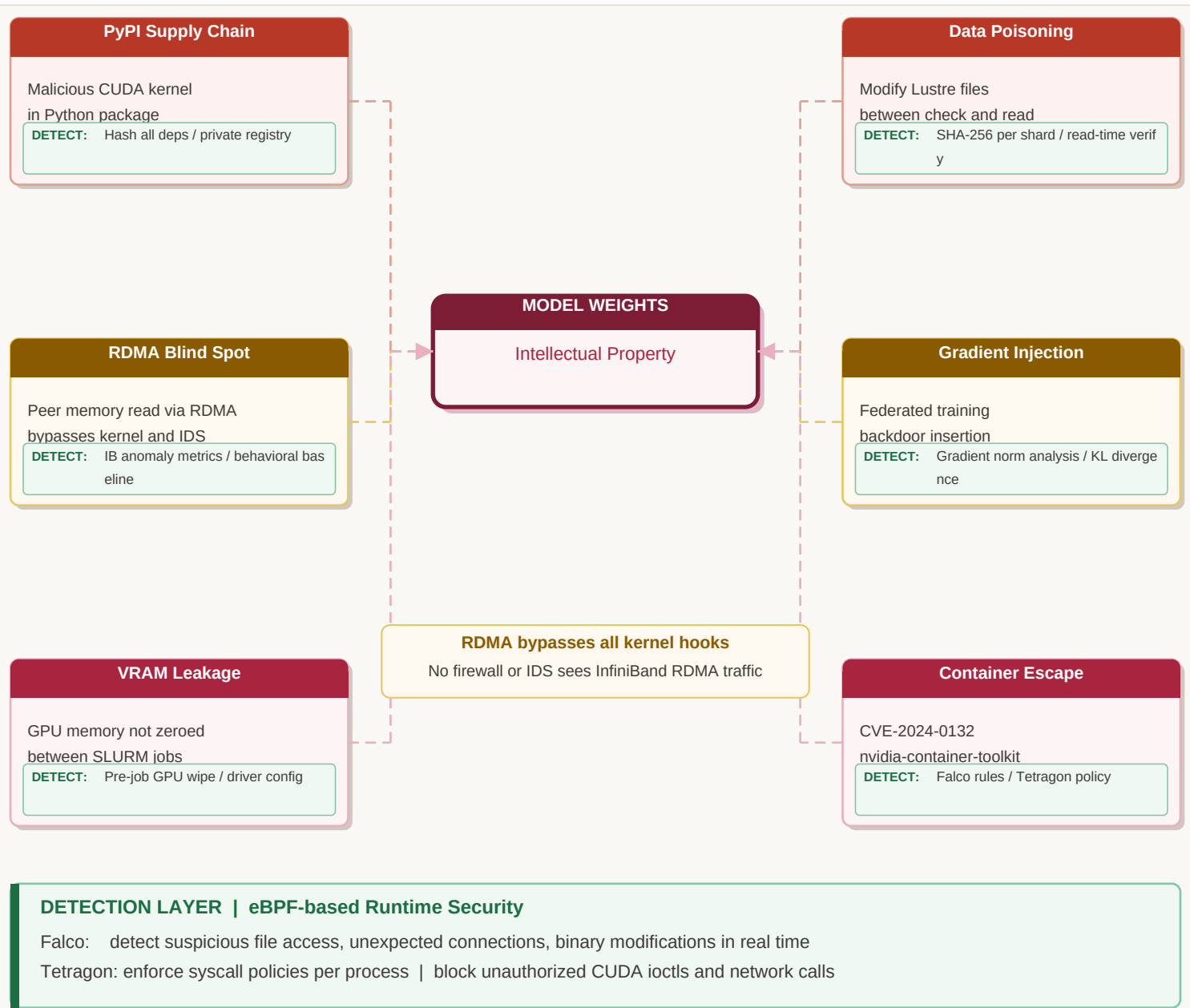


Diagram 4 | Security Threat Surface

Attack vectors specific to HPC AI workloads and detection methods



Threat severity: Critical High Medium